

TASK-1:

```
!pip install sumy
```

```
Collecting sumy
  Downloading sumy-0.12.0-py3-none-any.whl.metadata (8.3 kB)
Collecting breadability>=0.1.20 (from sumy)
  Downloading breadability-0.1.20.tar.gz (32 kB)
  Preparing metadata (setup.py) ... done
Collecting docopt-ng>=0.6.1 (from sumy)
  Downloading docopt_ng-0.9.0-py3-none-any.whl.metadata (13 kB)
Collecting lxml-html-clean (from sumy)
  Downloading lxml_html_clean-0.4.5-py3-none-any.whl.metadata (2.4 kB)
Requirement already satisfied: nltk>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from sumy) (3.9.1)
Collecting pycountry>=18.2.23 (from sumy)
  Downloading pycountry-26.2.16-py3-none-any.whl.metadata (12 kB)
Requirement already satisfied: requests>=2.7.0 in /usr/local/lib/python3.12/dist-packages (from sumy) (2.32.4)
Requirement already satisfied: setuptools>=65.0.0 in /usr/local/lib/python3.12/dist-packages (from sumy) (75.2.0)
Collecting docopt<0.7,>=0.6.1 (from breadability>=0.1.20->sumy)
  Downloading docopt-0.6.2.tar.gz (25 kB)
  Preparing metadata (setup.py) ... done
Requirement already satisfied: chardet in /usr/local/lib/python3.12/dist-packages (from breadability>=0.1.20->sumy) (5.2.0)
Requirement already satisfied: lxml>=2.0 in /usr/local/lib/python3.12/dist-packages (from breadability>=0.1.20->sumy) (6.1.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk>=3.0.2->sumy) (8.4.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk>=3.0.2->sumy) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk>=3.0.2->sumy) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk>=3.0.2->sumy) (4.67.3)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.7.0->sumy) (3.4.1)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.7.0->sumy) (3.18)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.7.0->sumy) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.7.0->sumy) (2025.11.12)
Downloading sumy-0.12.0-py3-none-any.whl (73 kB)
73.5/73.5 kB 3.5 MB/s eta 0:00:00
Downloading docopt_ng-0.9.0-py3-none-any.whl (16 kB)
Downloading pycountry-26.2.16-py3-none-any.whl (8.0 MB)
8.0/8.0 MB 68.7 MB/s eta 0:00:00
Downloading lxml_html_clean-0.4.5-py3-none-any.whl (14 kB)
Building wheels for collected packages: breadability, docopt
  Building wheel for breadability (setup.py) ... done
  Created wheel for breadability: filename=breadability-0.1.20-py2.py3-none-any.whl size=21695 sha256=6ec17207d8d97531e689df
  Stored in directory: /root/.cache/pip/wheels/32/99/64/59305409cacd03aa03ebddfb31a9db34b1fa7033bd41972662
  Building wheel for docopt (setup.py) ... done
  Created wheel for docopt: filename=docopt-0.6.2-py2.py3-none-any.whl size=13706 sha256=896c199c90deb0da1dbdc5ba83c01c2de33
  Stored in directory: /root/.cache/pip/wheels/1a/bf/a1/4cee4f7678c68c5875ca89eaccf460593539805c3906722228
Successfully built breadability docopt
Installing collected packages: docopt, pycountry, lxml-html-clean, docopt-ng, breadability, sumy
Successfully installed breadability-0.1.20 docopt-0.6.2 docopt-ng-0.9.0 lxml-html-clean-0.4.5 pycountry-26.2.16 sumy-0.12.0
```

```
import nltk
nltk.download('punkt')
nltk.download('punkt_tab')

from sumy.parsers.plaintext import PlaintextParser
from sumy.nlp.tokenizers import Tokenizer
from sumy.summarizers.lsa import LsaSummarizer

text = """
Good morning everyone. I am Nandini. I am pursuing my 3rd year of B.Tech in Artificial Intelligence and Machine Learning at
"""

parser = PlaintextParser.from_string(text, Tokenizer('english'))
summarizer = LsaSummarizer()

summary = summarizer(parser.document, 2) # pick top 2 sentences

for sentence in summary:
    print(sentence)

I have knowledge of Python, Machine Learning, and Natural Language Processing.
I have worked on projects such as a Student Internship Application Portal.
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
```

TASK-2:

```
!pip install transformers torch
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.10.2)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.11.0+cpu)
```

Requirement already satisfied: huggingface-hub<2.0,>=1.5.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.5.0)

Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.2)

Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)

Requirement already satisfied: regex>=2025.10.22 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)

Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.23.0)

Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from transformers) (0.25.1)

Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.8.0)

Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)

Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.29.2)

Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)

Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)

Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)

Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)

Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)

Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)

Requirement already satisfied: click>=8.4.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (8.1.8)

Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (1.2.0)

Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (0.23.0)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.37.0)

Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (1.5.3)

Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (13.9.4)

Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (0.0.2)

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch) (3.0.3)

Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (4.7.0)

Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (2025.11.11)

Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (1.0.7)

Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (3.10.1)

Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (0.14.0)

Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer->transformers) (3.0.0)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer->transformers) (2.19.2)

Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer->transformers) (0.1.2)

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM
```

```
article = ""
```

```
Good morning everyone. My name is Nandini. I am pursuing my 3rd year of B.Tech in Artificial Intelligence and Machine Learning at KSRM College of Engineering.
```

```
tokenizer = AutoTokenizer.from_pretrained("sshleifer/distilbart-cnn-12-6")
```

```
model = AutoModelForSeq2SeqLM.from_pretrained("sshleifer/distilbart-cnn-12-6")
```

```
inputs = tokenizer(article, return_tensors="pt", truncation=True, max_length=1024)
```

```
summary_ids = model.generate(
    inputs["input_ids"],
    num_beams=4,
    max_length=60,
    min_length=20,
    early_stopping=True
)
```

```
print(tokenizer.decode(summary_ids[0], skip_special_tokens=True))
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:112: UserWarning:
```

```
The secret `HF_TOKEN` does not exist in your Colab secrets.
```

```
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as an environment variable, and restart this notebook. You will be able to reuse this secret in all of your notebooks.
```

```
Please note that authentication is recommended but still optional to access public models or datasets.
```

```
warnings.warn(
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast downloads.
```

```
config.json: 100% 1.80k/1.80k [00:00<00:00, 36.3kB/s]
```

```
tokenizer_config.json: 100% 26.0/26.0 [00:00<00:00, 390B/s]
```

```
vocab.json: 100% 899k/899k [00:00<00:00, 9.31MB/s]
```

```
merges.txt: 100% 456k/456k [00:00<00:00, 6.36MB/s]
```

```
pytorch_model.bin: 100% 1.22G/1.22G [00:14<00:00, 116MB/s]
```

```
model.safetensors: 100% 1.22G/1.22G [00:18<00:00, 64.7MB/s]
```

```
[transformers] Please make sure the generation config includes `forced_bos_token_id=0`.
```

```
Loading weights: 100% 358/358 [00:00<00:00, 768.81it/s]
```

```
Nandini is pursuing her 3rd year of B.Tech in Artificial Intelligence and Machine Learning at KSRM College of Engineering.
```

TASK-3:

Feature	Extractive Summarization	Abstractive Summarization
Definition	Selects and combines important sentences directly from the original text.	Generates new sentences that capture the meaning of the original text.
Output	Uses the exact words and phrases present in the source.	Uses paraphrasing and may introduce new words not present in the source.

Feature	Extractive Summarization	Abstractive Summarization
Example	Original: "Nandini is a third-year AIML student. She knows Python and NLP." Summary: "Nandini is a third-year AIML student. She knows Python and NLP."	Original: "Nandini is a third-year AIML student. She knows Python and NLP." Summary: "An AIML student skilled in Python and NLP."

TASK-4:

```
import spacy

nlp = spacy.load('en_core_web_sm')

text = '''
Riya went on a special date with Arjun on Saturday evening.
They met at Moonlight Cafe in Hyderabad and enjoyed their favorite food.
Later, they watched a movie and spent quality time together.
Riya considered it one of the happiest days of her life.
'''

doc = nlp(text)

for ent in doc.ents:
    print(ent.text, '->', ent.label_)
```

```
Riya -> PERSON
Arjun -> PERSON
Saturday -> DATE
evening -> TIME
Moonlight Cafe -> FAC
Hyderabad -> GPE
Riya -> PERSON
the happiest days -> DATE
```

TASK-5:

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

# Telugu Translation
tokenizer = AutoTokenizer.from_pretrained("Helsinki-NLP/opus-mt-en-mul")
model = AutoModelForSeq2SeqLM.from_pretrained("Helsinki-NLP/opus-mt-en-mul")

text_to_translate = "Technology is changing the world and helping people learn new skills every day."

# Translate to Telugu
inputs = tokenizer(">>tel<< " + text_to_translate, return_tensors="pt")
translated_ids = model.generate(**inputs)
telugu_text = tokenizer.decode(translated_ids[0], skip_special_tokens=True)

print("Telugu Translation:")
print(telugu_text)

# Save Telugu output to text file
with open("telugu_output.txt", "w", encoding="utf-8") as f:
    f.write(telugu_text)

# Translate to Kannada
inputs = tokenizer(">>kan<< " + text_to_translate, return_tensors="pt")
translated_ids = model.generate(**inputs)
kannada_text = tokenizer.decode(translated_ids[0], skip_special_tokens=True)

print("\nKannada Translation:")
print(kannada_text)

# Save Kannada output to text file
with open("kannada_output.txt", "w", encoding="utf-8") as f:
    f.write(kannada_text)
```

```

config.json: 100%                               1.40k/1.40k [00:00<00:00, 29.6kB/s]
tokenizer_config.json: 100%                      44.0/44.0 [00:00<00:00, 2.04kB/s]
source.spm: 100%                                790k/790k [00:00<00:00, 9.42MB/s]
target.spm: 100%                                707k/707k [00:00<00:00, 15.5MB/s]
vocab.json: 100%                                1.42M/1.42M [00:00<00:00, 20.7MB/s]
/usr/local/lib/python3.12/dist-packages/transformers/models/arian/tokenization_arian.py:176: UserWarning: Recommended: pip
  warnings.warn("Recommended: pip install sacremoses.")
pytorch_model.bin: 100%                          310M/310M [00:05<00:00, 98.7MB/s]
model.safetensors: 100%                         310M/310M [00:08<00:00, 33.9MB/s]
Loading weights: 100%                           258/258 [00:00<00:00, 5672.94it/s]
generation_config.json: 100%                     293/293 [00:00<00:00, 26.5kB/s]
Telugu Translation:
ప్రపంచాన్ని మార్చుకోవడం, ప్రతిరోజు కొత్త సామర్థ్యాలను నేర్చుకోవడానికి ప్రజలకు సహాయం చేస్తుంది.
Kannada Translation:
ಲೋಕವನ್ನು ಬದಲಾಯಿಸುತ್ತಿದೆ ಮತ್ತು ಪ್ರತಿ ದಿನ ಹೊಸ ಕೌಶಲಗಳನ್ನು ಕಲಿಯಲು ಜನರಿಗೆ ಸಹಾಯಮಾಡುತ್ತದೆ.

```

TASK-5:

```

from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

# Load multilingual model
model_name = "facebook/nllb-200-distilled-600M"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSeq2SeqLM.from_pretrained(model_name)

# Input text
text_to_translate = "Good night, how are you?"

# Set source language
src_lang = "eng_Latn"
tokenizer.src_lang = src_lang # This tells the tokenizer the source language

# Tokenize input (without source language prefix in the text itself)
inputs = tokenizer(text_to_translate, return_tensors="pt")

# Translate to Telugu
telugu_ids = model.generate(
    **inputs,
    forced_bos_token_id=tokenizer.convert_tokens_to_ids("tel_Telu") # Changed from __tel_Telu__
)

telugu_text = tokenizer.batch_decode(
    telugu_ids, skip_special_tokens=True
)[0]

print("Telugu:", telugu_text)

# Save Telugu output
with open("telugu_output.txt", "w", encoding="utf-8") as f:
    f.write(telugu_text)

# Translate to Kannada
kannada_ids = model.generate(
    **inputs,
    forced_bos_token_id=tokenizer.convert_tokens_to_ids("kan_Knda") # Changed from __kan_Knda__
)

kannada_text = tokenizer.batch_decode(
    kannada_ids, skip_special_tokens=True
)[0]

print("Kannada:", kannada_text)

# Save Kannada output
with open("kannada_output.txt", "w", encoding="utf-8") as f:
    f.write(kannada_text)

```

```

Loading weights: 100%                               512/512 [00:00<00:00, 7128.36it/s]
Telugu: గుడ్ నైట్, మీరు ఎలా ఉన్నారు?
Kannada: ಗುಡ್ ನೈಟ್, ನೀವು ಹೇಗೆ?

```

TASK-6:

```

from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

# Load multilingual model
model_name = "facebook/nllb-200-distilled-600M"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSeq2SeqLM.from_pretrained(model_name)

# FAQ database
faq = {
    "What is AI?": "Artificial Intelligence is the simulation of human intelligence by machines.",
    "What is Machine Learning?": "Machine Learning is a branch of AI that enables systems to learn from data.",
    "Who are you?": "I am a simple multilingual FAQ bot."
}

# User input
question = "What is AI?"
language = "tel_Telu" # Use "kan_Knda" for Kannada

# Get answer
answer = faq.get(question, "Sorry, I don't know the answer.")

# Set source language
tokenizer.src_lang = "eng_Latn"

# Translate answer
inputs = tokenizer(answer, return_tensors="pt")
translated_ids = model.generate(
    **inputs,
    forced_bos_token_id=tokenizer.convert_tokens_to_ids(language)
)

translated_answer = tokenizer.decode(translated_ids[0], skip_special_tokens=True)

# Print answer
print("Question:", question)
print("Answer:", translated_answer)

```

Loading weights: 100%

512/512 [00:00<00:00, 5566.27it/s]

Question: What is AI?

Answer: కృత్రిమ మేధస్సు అనేది యంత్రాలచే మానవ మేధస్సును అనుకరించడం.

TASK-7:

```

articles = [
    """Apple unveiled its newest iPhone at an event in Cupertino on Tuesday. CEO Tim Cook said the device features a faster chip and improved camera. The phone will be available in stores starting next month, the company said, with prices starting at $999.""" ,
    """The World Health Organization announced new guidelines on Monday aimed at reducing sugar consumption worldwide. Officials in Geneva said the changes could prevent millions of cases of diabetes over the next decade if adopted by national governments.""" ,
    """Tesla reported record deliveries for the third quarter, beating analyst expectations. The Texas-based company said demand remained strong in China and Europe despite increased competition from traditional automakers entering the EV market.""" ,
    """NASA's Perseverance rover discovered new evidence of ancient water activity on Mars, according to a paper published this week. Scientists at the Jet Propulsion Laboratory said the findings strengthen the case that the planet once had a wetter climate.""" ,
    """Manchester United announced the signing of a new midfielder from Barcelona for a reported fee of 80 million euros. The 24-year-old player will join the club ahead of the new season starting in August.""" ,
]

```

```

import spacy
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

# Load each model ONCE – reused for every article
nlp = spacy.load('en_core_web_sm')

# Direct loading for summarization model
summarizer_tokenizer = AutoTokenizer.from_pretrained("sshleifer/distilbart-cnn-12-6")
summarizer_model = AutoModelForSeq2SeqLM.from_pretrained("sshleifer/distilbart-cnn-12-6")

```

Loading weights: 100%

358/358 [00:00<00:00, 8073.34it/s]

```

def digest_article(text):
    # Summarise (abstractive, short)
    inputs = summarizer_tokenizer(text, return_tensors="pt", truncation=True, max_length=1024)
    summary_ids = summarizer_model.generate(inputs["input_ids"], num_beams=4, max_length=40, min_length=10, do_sample=False)
    summary_text = summarizer_tokenizer.decode(summary_ids[0], skip_special_tokens=True)

    # Extract entities
    doc = nlp(text)
    entities = {}

    for ent in doc.ents:
        entities.setdefault(ent.label_, set()).add(ent.text)

    return summary_text, entities

for i, article in enumerate(articles, 1):
    summary, entities = digest_article(article)

    print(f"--- Article {i} ---")
    print("Summary :", summary)
    print("Entities:")

    for label, names in entities.items():
        print(f"  {label}: {' '.join(names)}")

    print()

```

```

--- Article 1 ---
Summary : CEO Tim Cook said the device features a faster chip and improved camera . The phone will be available in stores s
Entities:
  ORG: Apple
  GPE: Cupertino
  DATE: next month, Tuesday
  PERSON: Tim Cook
  MONEY: 999

--- Article 2 ---
Summary : The World Health Organization announced new guidelines aimed at reducing sugar consumption worldwide . Officials
Entities:
  ORG: The World Health Organization
  DATE: the next
decade, Monday
  GPE: Geneva
  CARDINAL: millions

--- Article 3 ---
Summary : Tesla reported record deliveries for the third quarter, beating analyst expectations . The Texas-based company se
Entities:
  DATE: the third quarter
  GPE: China, Texas
  LOC: Europe
  ORG: EV

--- Article 4 ---
Summary : NASA's Perseverance rover discovered new evidence of ancient water activity on Mars . Scientists said the finding
Entities:
  ORG: NASA, the Jet Propulsion Laboratory
  LOC: Mars
  DATE: this week

--- Article 5 ---
Summary : Manchester United announce signing of Barcelona midfielder from Barcelona . The 24-year-old midfielder will join
Entities:
  PERSON: Manchester United
  GPE: Barcelona
  CARDINAL: 80 million
  DATE: August, 24-year-old

```